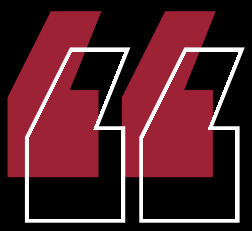


PROFESSIONAL EDUCATION

Digital Plus+ Programs

Blockchain:
Disruptive Technology



Come with us on a journey to redefine the business world

Dr. Abel Sanchez

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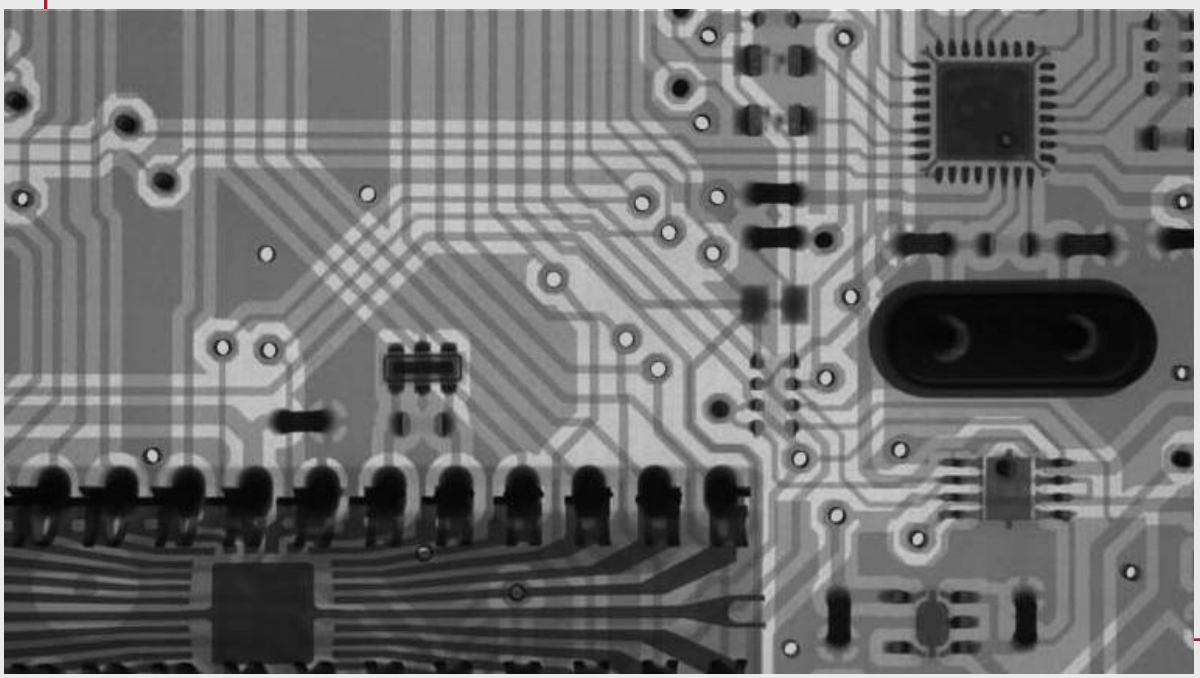
20 Benefits of Joining the Alumni Community

Technology

Blockchain has enjoyed runaway popularity in the technological world over the past few years. A potent force **free from geographic and economic barriers**, Blockchain has thoroughly disrupted our accepted ways of doing business. And, more importantly, **it's here to stay.**

As one of the **most important drivers of digital transformation**, Blockchain will undoubtedly have a **profound impact on job markets, organizational culture, governments and corporate strategy.**

1. **Encryption**
2. **Blockchains**
3. **Ethereum**
4. **Crypto currencies**
5. **Smart contracts**
6. **Digital signatures**



Impact of Blockchain

60%

of CIOs were about to integrate blockchain into their infrastructure by the end of 2020

Source: MVP

53%

of C-Suite professionals identified blockchain as a crucial part of their organizationalinfrastructure by 2020.

Source: Deloitte

77%

of executives agree that they will lose their competitive advantage if they do not adopt blockchain.

Source: Deloitte

The “Fourth Generation”

Blockchain connects decentralized systems in a format that is open, versatile and unalterably secure without relying on third parties. That way, all of us can participate directly in the viral change that is sweeping the business world

That is why MIT Professional Education is proud to announce its new online course **Blockchain: Disruptive Technology.**

This course teaches professionals from diverse sectors about the **fundamentals of Blockchain technology and its practical applications.** The goal is not only to understand how this technology works, but also to learn **how to apply it for maximum benefit** in digital transformation projects



About this Course



This course provides professionals from various sectors with an insight into the **fundamentals of blockchain and its practical applications.**

The objective is not only to **understand how this technology works**, but also to learn what its **applications** are in order to obtain the **maximum benefit in the digital transformation** projects that are developed.

8 weeks | 8-10 hours a week | Online

Key Takeaways

This course takes a holistic perspective on Blockchain and uses theoretical and practical tools to address the history and fundamentals of Blockchain. It also focusses on current and future applications and related technologies.

The course emphasizes hands-on experience that is relevant for jobs and projects across multiple industries. After completing this course, you will:

01 **Understand the historical, technological, and commercial forces** that have led to the emergence of Blockchain.

02 **Analyze Blockchain's impact across industries**, and of course, in your own business.

03 **Master the fundamentals of Blockchain technology,** including distributed ledgers, blocks, chains, encryption, PKIs (public and private key infrastructure), hashing and consensus mechanisms: proof-of-work (PoW) and proof-of-stake (PoS).

04 Learn about the Blockchain Ethereum network and its potential applications.

05 **Become familiar with the growing range of Blockchain applications,** including bitcoin and other cryptocurrencies, intelligent contracts and digital signatures.

06 **Study cryptocurrency in-depth,** its strengths and weaknesses in the digital world, and how it will revolutionize e-commerce, finance and even legislation in many countries.

07 **Become aware of the deficiencies of previous web platforms** with respect to accessibility, monopolies, privacy, censorship, transparency and integrity and how Blockchain promises to resolve these issues.

08 **Become a catalyst for Blockchain technology** in your organization



Participant Profile

The online course **Blockchain: Disruptive Technology** is **applicable to all industries and sectors**, as well as for many distinct professional roles and companies. We have all felt Blockchain's disruptive impact in the rapid digital transformation of the business world, which is why this course analyzes a wellrounded selection of technical applications, cases and tools.

CTOs, CINOs and leaders of technological departments or projects who need to understand the new possibilities that Blockchain offers for digital transformation and advancements in technical areas.

Technical professionals who want to obtain a broader strategic vision about how Blockchain can transform their business operations.

CEOs, entrepreneurs and functional managers in technical areas or sectors who want to create new business models and upscale their current operations.

Professionals who want to understand the benefits and opportunities that Blockchain offers for digital transformation in their organization.

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Our Professional Alumni Experience

Jorge Valdivia
Director at Fiserv

A better understanding of this new technology positions me in a better place to confidently talk to clients and/or co-workers about blockchain and identify better potential use cases to leverage for adoption.

Dmitry Valkov
Inspections Leader at Public Company Accounting Oversight Board

Very relevant and insightful content. I obtained an understanding of a much broader picture of what blockchain technology is and its various use cases outside of crypto. I feel confident at work with the newly acquired knowledge.

Ludivine Gustave Dit Duflo
Founder of Madizy Management

With this newfound knowledge, I can bring innovative ideas to my team and projects, proposing blockchain technology to streamline financial transactions and reduce costs. I can develop new products or services that leverage blockchain technology to provide greater value to my customers.

Meet our Instructors

* Speakers and topics are subject to change without notice.



Dr. Abel Sanchez

Executive Director of MIT's Geospatial Data Center, Research Scientist; Center for Complex Engineering Systems, Sociotechnical Systems Research Center, under the Schwarzman School of Computer Science

Dr. Sanchez holds a Ph.D. from the Massachusetts Institute of Technology (MIT). He is the Executive Director of MIT's Geospatial Data Center (GDC). His areas of specialty include the Internet of Things (IOT), Big Data, Cybersecurity, and Digital Innovation. He teaches graduate courses in Data Science, Cybersecurity, and Innovation. For the past eight years his research has focused on architecting large-scale computation.

In IOT, Dr. Sanchez led the global network architecture for the Internet of Things at MIT. The design addresses large-scale computation. Compared to the largest numbers in the world, 2 billion computers, 7 billion phones, and 7 billion people, IOT is orders of magnitude bigger. In similar work, Dr. Sanchez directed simulation of the U.S. critical infrastructure with the National Infrastructure Simulation and Analysis Center (NISAC).ia.

In data science, early work included supply chain information engineering, analytics, simulation, and visualization with Wal-Mart, Kraft, and SAP. Dr. Sanchez extended his work to global anti-counterfeiting efforts with Johnson & Johnson, SAP, and Altria. Helping Altria scale track-and-trace using the Electronic Product Code Information Services standard from RFID. The work produced a global infrastructure used by industry and government to this day.

In enterprise computing, Dr. Sanchez led the design of a global data infrastructure simulator, modeling follow-the-sun engineering, to evaluate the impact of competing architectures on the performance, availability and reliability of the system for Ford Motor Company. The simulator modeled user actions, applications, background processes, network load, servers, storage, and global data centers. The work identified data center reductions opportunities estimated at a billion dollars in savings.

In cyber security, Dr. Sanchez directed impact analysis of large-scale cyber-attacks designing Cyber Ranges for the Department of Defense (DOD). Conducting repeatable experiments in impact analysis and the ability to model the cyber environment in a highly portable fashion. Looking at the insider threat Dr. Sanchez led the DOD security study on Enterprise Resource Planning systems across the United States Armed Forces.

In password security, Dr. Sanchez led the design of a password firewall (negative authentication) for the Intelligence Advanced Research Projects Activity (IARPA) agency. The Negative Filtering or Negative Authentication (NA) approach utilizes a form of complement profiles which resembles the censoring and maturation process of T- cells in the immune system.

In machine learning, addressing financial fraud, Dr. Sanchez designed a situational awareness framework that exploits different perspectives of the same financial data and assigns risk scores to entities (e.g. payment documents) to improve false positive ratios and assist the identification of fraudulent activity in huge and unlabeled financial data in collaboration with Accenture.

In physical security, Dr. Sanchez is developing algorithms to assess risk in the integration of information technology (IT) and operations technology (OT).
Dr. Sanchez is the founder and Chief Software Architect of the Open Source RFID platform project. Dr. Sanchez' software systems are used by Samsung, NEC, NTT, Hitachi, Motorola, SAP, IBM, and Microsoft. Other software initiatives are in use by Sandia National Laboratories, MIT, and by several organizations in East Asia and Europe.
Dr. Sanchez advises companies in the Americas, Europe, Middle East, and Asia.

> "We cannot bet on the artificial intelligence, or the human being separate. We must get the fusion of both."

Affiliations

- MIT Geospatial Data Center (GDC)
- MIT Sociotechnical Systems Research Center (SSRC)
- MIT Institute for Data, Systems, and Society (IDSS)
- MIT Center for Complex Engineering Systems (CCES)
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT Schwarzman School of Computer Science
- MIT AutoID Laboratory

Areas of Interest and Expertise

- Machine Learning
- Cyber/Physical Security
- Enterprise Computing
- Data Science
- IOT

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation´s Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity



Prof. John R. Williams

Professor of Information Engineering, Civil and Environmental Engineering and Director of MIT Geospatial Data Center, and a faculty member in the Center for Computational Science and Engineering part of the Schwarzman School of Computer Science

John Williams holds a BA in Physics from Oxford University, a MS in Physics from UCLA, and a Ph.D. in Numerical Methods from University of Wales, Swansea. His research focuses on the application of Exascale computation to problems in cyber-physical systems, security and energy. His work on fault-tolerant computing using container migration won the IEEE, High Performance Extreme Computing Conference award for Best Innovation in 2019. He is director of MIT's Geospatial Data Center and from 2006-2012, was Director of the MIT Auto-ID Laboratory, where the Internet of Things was invented.

He is author or co-author of over 250 journal and conference papers, as well as the books on Rock Mechanics and RFID Technology.

He contributed to the 2013 report for the UK Office for Science Foresight Project- The Future of Manufacturing. Alongside Bill Gates and Larry Ellison, he was named as one of the 50 most powerful people in Computer Networks. He consults to companies including Accenture, Schlumberger, Shell, Total, Exxon, SAP Research, Microsoft Research, Kajima Corp, US Lincoln Laboratory, Sandia National Laboratories, US Intelligence Advanced Research Projects Activity, Motorola, Phillip-Morris Inc., Ford Motor Company, Exxon-Mobil, Shell, Total, and ARAMCO.

His international collaborations include Oxford and Cambridge Universities, HKUST, KACST, Alfaisal University, PolyU Hong Kong, Imperial College of Science and Technology UK, Malaysia University of Science and Technology (MUST), and Masdar Institute of Science and Technology Abu Dhabi. He organized the first Cyber-Physical Security Conference in the UK (2011) and along with Dr. Sanchez, he runs the MIT Applied Cyber Security Professional Education summer course. At MIT he teaches courses Architecting Software Systems (MIT 1.125) and Engineering Computation and Data Science (MIT 1.00/1.001).

In data engineering and data science, early work included simulation of Ford's global network, analysis of SAP smart grid billing system. For Altria, he analyzed the performance of item level tagging and also their implementation of an anti-counterfeiting system using the Electronic Product Code (EPC).

In password security, Dr. Williams was PI that developed the algorithms for a negative password authentication system for the Intelligence Advanced Research Projects Activity (IARPA) agency.

Dr. Williams advises companies in the Americas, Europe, Middle East, and Asia.

> "Once you climb that first mountain of knowledge, you will be able to teach yourself many things."

Affiliations

- MIT Department of Civil and Environmental Engineering
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Schwarzman School of Computer Science
- MIT Geospatial Data Center (GDC)
- MIT Auto-ID Laboratory
- MIT Center for Complex Engineering Systems (CCES)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT System Design and Management Program

Areas of Interest and Expertise

- Information Technology,
- Cyber/Physical Security
- Web-Based Education Technology
- Large Scale Network Simulation
- GeoNumerics of Granular and Powder Systems
- Modern Software Architecting and Cyber Security
- Web Services and Distributed Computing
- Discrete Element Simulation and Analysis of Discontinua

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation´s Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity

Course Outline

Module 0
Introduction

Module 1
Integration

1. Data Integration
2. Service Integration
3. Microservices Evolution
4. Integration Evolution
5. Blockchain Integration

Module 2
Visual Blockchain

1. Fundamentals
2. Coinbase
3. Keys

Module 3
Tokenization, Currencies and Contracts

1. Transformation
2. Tokenization
3. Central Banks
4. Tokenization Platforms
5. Transactions
6. Private Blockchain
7. Blockchain Web Clients – Customized DApps
8. Blockchain Education Certificates
9. Private-Public Partnerships

Module 4
Exchanges

1. Centralized vs. Decentralized Exchanges
2. 0x Protocol
3. 0x Technical Implementation

Module 5
Governance

1. Why Do We Need Governance?
2. Governance Examples
3. Centralized or Distributed?
4. The Need
5. Considerations

Module 6
Consortia

1. Types of Consortia
2. Going Further
3. More than a Distributed Database
4. People
5. Consortium Boom
6. Startup
7. Islands of Trust
8. How to start

Module 7
Cloud

1. Drivers
2. First Project
3. Codebases
4. Serverless
5. Ledgers
6. Reinvention
7. Tools and Platforms

Module 8
Emerging Technologies

1. Stablecoins
2. NFTs
3. Virtual Property and Metaverse
4. Smart Contracts
5. VR and A

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Certificate of Completion

All participants who successfully complete **Blockchain: Disruptive Technology** will receive an MIT Professional Education Certificate of Completion. In addition, they will also earn 6.4 Continuing Education Units (MIT CEUs).

In order to obtain CEUs, participants must complete a required CEU accreditation form. CEUs are calculated based on the number of learning hours in each course.



*The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

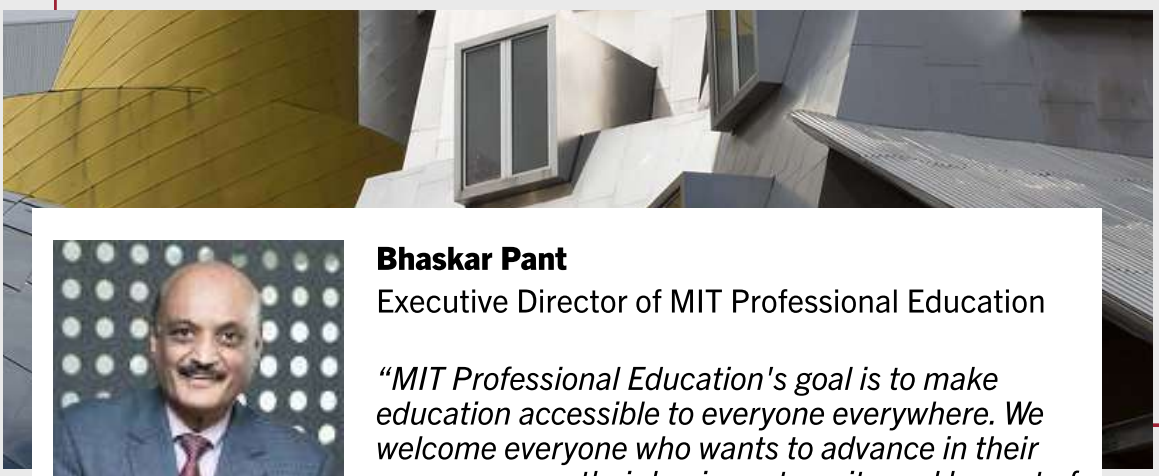
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**





Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:
Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:
Available in Open Enrollment courses with a minimum of participants.

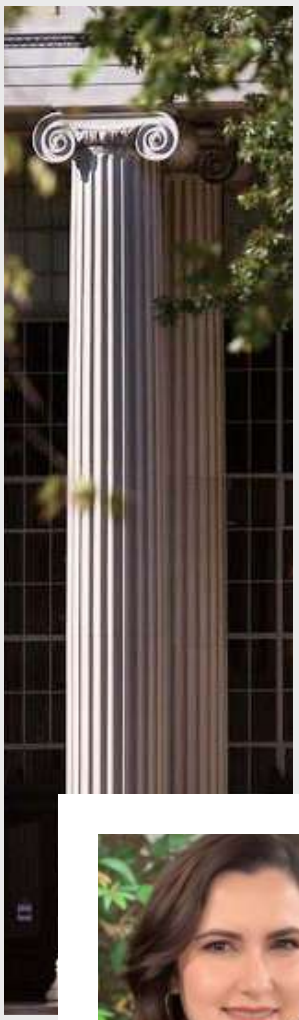


For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience.**

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology.You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto
Director of Global Programs at MIT Professional Education

"Blockchain technology is reinventing and facilitating a multitude of online operations. Blockchain increases trust, security, transparency, and the traceability of shared data across a network with cost-saving efficiencies. Understanding blockchain is an accelerating necessity for professionals to maintain a competitive edge in the Fourth Industrial Revolution."

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

1. **15% discount** for MIT Professional Education online courses and short-duration, in-person programs.
2. **Continued access to the virtual campus for six months after completion** of the course.
3. **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network.**
4. Invitation to the **MIT Professional Education Alumni LinkedIn group.**
5. **Networking opportunities** with other MIT Professional Education Alumni.

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PROFESSIONAL EDUCATION
Digital Plus+ Programs

Are you ready to become a leader in the latest technologies?

Frequently Asked Questions

global alumni
-excellence or nothing-

For more information, please contact us:

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Email: **professionalprograms@mit.edu**

Website: **https://professionalprograms.mit.edu**